

Vokr — Zero-Cost Production Tech Stack & Production Readiness Checklist

Definitive implementation reference for the Vokr engineering team.

Version 1.0 · 7 September 2026 · Supersedes all prior stack documents and checklists.
Derived from: review of the 27-page `vokr-production` codebase, the original Vokr Tech Stack & Checklist, the backend engineering scope, provider free-tier verification, and capacity analysis.
Currency assumption: ₹88 = US\$1. USD figures are authoritative; rupee figures are indicative. Verify the current rate.

How to read the evidence tags in this document

VERIFIED	Confirmed against the provider's official documentation or pricing page on 7 Sep 2026. Treat as fact, but re-check before any major commitment — providers change quotas with notice.
ESTIMATE	Engineering judgement derived from stated assumptions. The assumptions are what you should argue with, not the number.
LOAD TEST	Cannot be known without measuring the real application. Do not plan around a guess.
UNVERIFIED	No first-party source obtained. Must be confirmed by the team before it is relied upon.

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1. Final Verified Production Tech Stack		
Layer	Component	Configuration / notes
Frontend + API	Next.js (App Router) + TypeScript	One deployable. Pages and API routes in a single app. Replaces the 27 standalone HTML files.
Hosting	Google Cloud Run — <code>asia-south1</code> (Mumbai)	<code>min-instances=0</code> , <code>max-instances=3</code> , 1 vCPU, 512 MiB, concurrency 80. Kept warm by a Cloud Scheduler ping.
Edge	Cloudflare Free	DNS, CDN, WAF, SSL, bot protection. Aggressive cache rules on marketing pages and static assets.
Database	Supabase Postgres — Mumbai region	Free plan (nano compute). Sole datastore: catalog, orders, inventory, carts, rate limits, ledger.
ORM	Prisma	Transaction-mode pooler with <code>pgbouncer=true</code> ; direct/session connection for the checkout transaction.
Auth	Supabase Auth	Email + Google. Guest checkout supported. Users live in your own Postgres. Custom SMTP via Brevo (mandatory).
Payments	Razorpay	Orders API, webhook signature verification, idempotency keys. UPI, cards, net banking, wallets.
Shipping / COD	Shiprocket	Manual panel at launch. API integration deferred until volume justifies the plan tier.
Outbound email	Brevo	Transactional email and Supabase Auth custom SMTP. Single sending domain.
Inbound email	Zoho Mail Free	5 mailboxes: support@, grievance@, privacy@, legal@, careers@.
Object storage	Cloudflare R2	Product images + database backups. Public bucket on a custom domain.
Search	Static JSON index served from the API	Single source of truth. No search engine, no Postgres FTS.
Cache	None (in-process catalog cache only)	No Redis. 6-SKU catalog held in Cloud Run memory with a short TTL.
Jobs	Cloud Scheduler — 3 jobs	(1) keep-warm ping, (2) 6-hourly <code>pg_dump</code> to R2, (3) inventory reservation expiry. No queue.
Monitoring	Sentry Developer + Cloud Logging	Weekly log export to R2 for DPDP retention.
Backups	Self-managed <code>pg_dump</code> → R2	Every 6 hours. Supabase Free has no backups and no PITR.
CI/CD	GitHub Actions → Artifact Registry → Cloud Run	Cleanup policy on the registry from day one.
Secrets	Google Secret Manager	Exactly 6 active versions. Loaded once at process boot.
Domain	Cloudflare Registrar — <code>vokr.shop</code>	At-cost. The only unavoidable fixed recurring cost.

Explicitly not in the stack at launch: Redis · a separate queue or worker · Algolia or any search engine · Postgres full-text search · SMS / phone OTP · Clerk · Neon · Cloudflare Pages as a separate deploy target · Cloud Storage (GCS) · min-instances on Cloud Run.

Each was considered and rejected for a stated reason. See sections 2 and 9.

2. Why Each Component Was Selected

2.1 Selected — and the concrete reason

Component	Reason for selection
Next.js, single app	The existing codebase duplicates ~80 KB of CSS/JS across all 27 HTML files, and the duplication has <i>already</i> caused a defect: four pages carry a 15-entry search index including a product with no page, while 23 carry 13 entries. One deployable removes the class of bug and eliminates CORS and cookie-domain problems between a static frontend and a separate API.
Cloud Run (Mumbai)	Scales to zero, no servers to patch, and Mumbai keeps the app in the same city as the customers. VERIFIED the Cloud Run free tier is <i>not</i> region-restricted — it is applied as a spending-based discount at Tier 1 pricing, and <code>asia-south1</code> is a Tier 1 region.
Cloudflare Free	DNS, global CDN, WAF, SSL and bot protection at no cost. Also the layer that keeps Cloud Run egress bills near zero, since Cloud Run has no free egress allowance outside North America.
Supabase Postgres	Chosen over Neon on two grounds. First, region: Neon's only Asia-Pacific regions are Singapore and Sydney; Supabase has Mumbai. Second, and decisive, Neon's free tier caps at 100 compute-hours per month as a hard cutoff that suspends the database — an always-warm instance burns roughly 180. Neon Free cannot run an always-on store; Supabase Free can. VERIFIED
Supabase Auth	Chosen over Clerk on architecture, not cost. Identity lives in your own Postgres, so <code>orders.user_id</code> is a real foreign key and DPDP erasure is one system rather than two. VERIFIED 50,000 MAU free. <i>Correction to an earlier draft:</i> Clerk is not expensive here — it raised its free allotment to 50,000 monthly retained users on 5 Feb 2026. The reason to prefer Supabase Auth is referential integrity, not price.
Razorpay	The standard for Indian D2C. VERIFIED ₹0 setup, ₹0 annual maintenance, no monthly minimum, no volume floor. Purely transactional. Deep UPI support.
Shiprocket	Without a courier there is no COD, no PIN-code serviceability, no tracking, no RTO handling and no return labels — four separate promises already written into <code>terms.html</code> . Manual panel operation at launch keeps the fixed cost at zero.
Brevo	Highest free transactional allowance with full API, SMTP and webhook access. Doubles as the Supabase Auth SMTP provider, so one vendor covers both paths.
Zoho Mail	The only genuinely free custom-domain business email. Brevo sends but does not receive, and the site publishes five <code>@vokr.shop</code> addresses including a statutory <code>grievance@</code> .
Cloudflare R2	Zero egress at any volume. The codebase currently hotlinks 84 images from <code>cdn.shopify.com</code> — a third party's CDN that can break without warning and is not yours to depend on.
Static JSON search	The catalogue is 6 SKUs and the existing "index" is 13 entries covering products <i>and</i> support pages. Postgres FTS solves a problem Vokr does not have; the real defect is that the index is duplicated 27 times and has drifted.
Cloud Scheduler	3 free jobs per billing account is exactly the number needed. VERIFIED
Sentry	Best-in-class TypeScript stack traces and source-map support at zero cost.

2.2 Rejected — and why

Rejected	Reason
Redis / Upstash	Postgres handles rate limiting (a table with an index), cart persistence, and inventory locking (<code>SELECT ... FOR UPDATE</code>). Upstash's free tier carries no uptime SLA and rate-limits at the budget cap; making the last-pair-in-stock guarantee depend on it invites overselling. One fewer vendor, one fewer network hop, one fewer free tier to breach.
Separate queue / worker	Three Cloud Scheduler jobs cover every scheduled need at this scale. Razorpay retries its own webhooks. Cloud Tasks' 1M free operations remain available if that changes.
Search engine / Postgres FTS	Over-specified for 13 searchable entries.
SMS / phone OTP at launch	TRAI's DLT framework requires Principal Entity registration at ~₹5,900 one-time plus 3–7 days of approval before a single message can be delivered. That is a poor first spend at low order volume. Deferred, with a required copy change — see item R14.
Cloud Run <code>min-instances=1</code>	Would cost roughly \$9.72/month for 1 vCPU / 512 MiB before free-tier offset. Unnecessary: the Cloud Run pricing page states that <i>idle instances that are not minimum instances are not charged</i> , so a 5-minute keep-warm ping delivers the same result for approximately nothing. VERIFIED
Google Cloud Storage for backups	The GCS free tier applies only in <code>us-east1</code> , <code>us-west1</code> and <code>us-central1</code> . R2 is free and has no egress charge. VERIFIED

3. Zero-Recurring-Cost Model

The honest headline: **literal ₹0/month is not achievable**. Infrastructure lands at roughly **₹250-400/month**, and the reason is mundane — a commercial storefront needs a domain.

3.1 Cost categories

Category	Amount	Detail
A. Truly ₹0 fixed	₹0	Cloud Run compute, Supabase Postgres + Auth, R2, Brevo, Zoho Mail, Sentry, Cloud Scheduler, Secret Manager, Artifact Registry, Cloud Logging, GitHub Actions, Cloudflare CDN/DNS/WAF/SSL, Razorpay account.
B. ₹0 within free-tier limits	₹0 → usage	Everything in category A. Each has a quota; section 4 lists them and the failure mode at the limit.
C. Transaction / business costs	Scales with revenue	Razorpay 2% + 18% GST = 2.36% effective . Shipping ~₹36-45/parcel. COD handling ~₹27 + 1.5% (min ~₹37). RTO on COD: 15-25% of COD orders, ~₹100-150 lost each . These are cost of goods sold, not infrastructure.
D. Mandatory paid	~₹221/mo + ~₹10-100/mo	Domain <code>vokr.shop</code> at ~\$30.20/year UNVERIFIED (third-party price list; confirm at checkout). Cloud Run egress from Mumbai at \$0.12/GiB with no free allowance outside North America VERIFIED .
E. Optional / deferred	₹0 for now	Supabase Pro (\$25), Brevo Starter (\$9), Sentry Team (\$26), Shiprocket API plan, DLT registration (₹5,900 one-time). All triggered by growth, none needed at launch.

Two structural facts that make "free" conditional.

- A Google Cloud billing account with a valid payment method is **mandatory** to access Always Free at all. **VERIFIED**
- GCP Free Tier limits are a **spending-based discount, not a spending ceiling**. They do not cap your bill. Unused limits do not roll over. This is why `max-instances` and billing budget alerts are launch blockers, not nice-to-haves.

3.2 Realistic monthly infrastructure budget

Scenario	Fixed infrastructure	Notes
Case A — testing, <10 orders/mo	~₹225/month	Domain ₹222 + ~₹2 egress. Everything else inside quota.
Case B — small real launch, 50-100 orders/mo	~₹240-740/month	Domain + ~₹16 egress. Upper bound only if Shiprocket API access is gated behind a paid plan UNVERIFIED .
Case C — early traction, 600-900 orders/mo	~₹5,500-6,300/month	Supabase Pro + Sentry Team + Shiprocket plan + egress. Against roughly ₹50-70 lakh GMV, infrastructure is under 0.15% of revenue.

Conclusion for planning: chasing the last ₹221 buys nothing. Budget ₹500/month for infrastructure, ignore it, and spend the attention on RTO instead — at 20% RTO the effective added cost per *delivered* COD order is around ₹62.50, which dwarfs the entire infrastructure bill within a few dozen orders.

4. Free-Tier Limits and Guardrails

Every limit below is **VERIFIED** against the provider's official documentation on 7 September 2026. The **failure mode** column is the one to internalise — some quotas degrade gracefully and some take the site down.

Service	Free allowance	Failure mode at the limit	Guardrail
Cloud Run compute	2M requests/mo; 180,000 vCPU-s; 360,000 GiB-s. Per billing account, resets monthly, no rollover.	Billed at standard rates. No service interruption.	<code>max-instances=3</code> . Billing budget alerts at \$1 / \$5 / \$20.
Cloud Run egress	1 GiB — North America only. Mumbai gets zero free egress.	Billed \$0.12/GiB. Never fails, only bills.	Cloudflare cache rules on all static assets and marketing pages. Images served from R2, not Cloud Run.
Supabase egress	5 GB/month uncached + 5 GB cached.	HTTP 402 across every Supabase service. The storefront goes down mid-trading-day with no graceful degradation. Sharpest risk in the stack.	Alert at 3 GB. Cache the catalog in memory. Never <code>SELECT *</code> . No images or logs in Supabase.
Supabase storage	500 MB database.	Writes fail. Cumulative — never resets.	Scheduled pruning of abandoned carts and rate-limit rows. Logs go to Cloud Logging, not Postgres.
Supabase compute	nano: shared CPU, ~500 MB RAM. 60 direct / 200 pooler connections.	Query latency degrades; connection errors under load.	Connections are not a concern at <code>max-instances=3</code> . Compute needs LOAD TEST .
Supabase backups	None. No backups, no PITR.	Total data loss on corruption.	6-hourly <code>pg_dump</code> to R2 with a tested restore. RPO = 6 hours.
Supabase Auth mailer	Built-in sender: 2 emails/hour, and only delivers to addresses on your own project team.	Silent failure. <code>signUp()</code> succeeds, the email never arrives, no client error. Breaks for every real customer.	Configure Brevo as custom SMTP before any real signup. Cap then starts at 30/hour and is adjustable in the dashboard.
Supabase Auth limits	50,000 MAU. Token-bucket rate limiting, max capacity 30; sign-in/sign-up 30 per 5 min per IP .	HTTP 429. If Auth is called server-side, every customer shares Cloud Run's egress IP and one bucket — an effective global cap of ~6 sign-ins/minute.	Forward the real client IP, or call Auth from the browser. Raise the email-send rate limit before launch.
Brevo	300 emails/day , shared across transactional and marketing. Full API/SMTP/webhooks on free.	Sending stops for the rest of the day. Order confirmations silently do not arrive. Daily cap — a spike day breaches it before the monthly average looks concerning.	Alert at 200/day. Note free-plan emails may carry Brevo branding UNVERIFIED — send yourself a test order confirmation.
Cloudflare R2	10 GB storage; 1M Class A (writes); 10M Class B (reads); egress always free at any volume.	Billed at \$0.015/GB-mo, \$4.50/M Class A, \$0.36/M Class B.	Retention policy on backup dumps. Not a realistic constraint otherwise.
Sentry	5,000 errors/mo; 1 user ; 30-day retention.	Events dropped.	Error-rate bound, not volume-bound. One broken route or crawler loop burns 5,000 in an afternoon. Enable spike protection; fingerprint aggressively.
Cloud Scheduler	3 jobs/month per billing account — not per project.	\$0.10/job/month.	All 3 allocated. A staging project cannot have its own three.
Artifact Registry	0.5 GB storage.	Billed silently.	Tightest quota in the stack. A Next.js image is 150–400 MB; three retained revisions exceeds it. Cleanup policy: keep 2 tags, delete untagged.
Secret Manager	6 active secret versions; 10,000 access operations/mo; 3 rotation notifications.	Billed at standard rates.	6 versions is exactly what Vokr needs — rotating any secret creates a seventh, so disable the old version rather than adding. Load at boot, never per request.
Cloud Logging	50 GiB ingestion per project/month. Default retention free.	Billed at standard rates.	Default retention is 30 days; DPDP Rules impose a one-year minimum on personal data, traffic data and processing logs. Export weekly to R2.
GitHub Actions	2,000 min/mo on private repos; unlimited on public. 500 MB Packages storage.	Hard stop — default Actions spending limit is \$0, so jobs fail rather than billing.	~400 builds/month at 5 min each. Not traffic-dependent.
Cloudflare Free	Unlimited static requests; 1,000 DNS records; 5 WAF custom rules.	Upgrade to Pro (\$20–25/mo).	5 custom WAF rules is the practical ceiling. Do not serve large media through the free CDN.
Razorpay	No volume limit. ₹0 setup, ₹0 AMC.	n/a	Live mode requires completed KYC — entity, PAN, GST, bank account. Start this early.

5. Practical Expected Capacity

5.1 Assumptions this section depends on

If these are wrong, every number below moves. They are the thing to challenge.

Assumption	Value	Basis
Conversion rate	2% of sessions	ESTIMATE Early D2C, mostly direct/referral. Cold paid traffic at 0.5–1% would double-to-quadruple pageviews per order.
Pages per session	~4	ESTIMATE
Pageviews per order	~200	Derived from the two rows above.
Origin requests per pageview	~3	ESTIMATE 1 HTML + 2 API; static assets cached at Cloudflare.
Emails per order	~4	ESTIMATE Confirmation, shipped, delivered, plus a share of auth emails.
Next.js SSR server time	~100 ms/request	LOAD TEST Could plausibly be 40 ms or 300 ms.
Origin egress per pageview	~50 KB gzipped	ESTIMATE Images served from R2.
Catalog cached in memory	Yes	Design decision. Changes Supabase egress headroom by roughly 4x. See 5.3.

5.2 Capacity by scale

Scale	Orders/day	Orders/month	Approx traffic	Free tier safe?	Recommendation
Very small	1–5	30–150	~6k–30k pageviews/mo	SAFE every quota under 10%	Launch here. Nothing to monitor except Sentry noise.
Small	5–25	150–750	~30k–150k pageviews/mo	SAFE largest usage ~30–60% (Supabase egress)	Add Brevo and Supabase egress dashboards. Comfortably free.
Growing	25–70	750–2,000	~150k–400k pageviews/mo	POSSIBLE BUT MONITOR	Alert at Brevo 200/day and Supabase 3 GB. Pre-authorise the Pro upgrade so on-call can act without a meeting.
Large	70+	2,000+	400k+ pageviews/mo	SHOULD UPGRADE Brevo breached; 402 risk real	Brevo Starter + Supabase Pro. Still no architecture change.

Dimension	Figure	Basis
Concurrent customers	~200–500 browsing	ESTIMATE LOAD TEST At <code>max-instances=3</code> . Raising it lifts this and costs compute beyond the free allowance.
Concurrent checkouts	Safe ~10–20 Monitor 20–50	LOAD TEST Bound by row-lock contention on a hot variant, not by throughput. Do not assume beyond 20 without measuring.
Peak / spike traffic	~150 orders in one day breaches Brevo	Daily and burst caps bite before monthly averages do. Plan launch day and festival sales separately — one month of Brevo Starter at \$9 is the whole fix.

5.3 The first bottleneck, in order

- Brevo, 300 emails/day → ~70–75 orders/day.** Hits first because it is a *daily* cap. Failure mode: confirmations silently stop. **VERIFIED** limit, **ESTIMATE** on emails per order.
- Supabase egress, 5 GB/month.** With the catalog queried per pageview (~20 KB), roughly **250,000 pageviews/month (~1,250 orders)**. With the catalog cached in memory (~5 KB/pageview average), roughly **1,000,000 pageviews/month**. Failure mode is worse than Brevo's: HTTP 402 across all services. **Plan for 250,000; treat more as earned by caching discipline.**
- Supabase storage, 500 MB → ~20,000–30,000 orders lifetime.** **ESTIMATE** Cumulative, never resets.
- Cloud Run compute → ~200,000–600,000 pageviews/month.** **LOAD TEST** vCPU-seconds bind before memory at 512 MiB.

Not bottlenecks at any realistic scale here: R2, Razorpay, Cloudflare, GitHub Actions, Cloud Run request count, Supabase MAU, Supabase connections. Sentry is bound by error rate, not order volume.

5.4 Technical capacity, ignoring free tiers

On paid tiers with the **identical architecture**: roughly **300–600 orders/day (10,000–18,000/month)** **ESTIMATE** before the first genuine architectural pressure appears — a background job runner for outbound email and courier API calls, because by then inline side effects hurt checkout latency and couple the order flow to third-party availability. Postgres itself would go considerably further, likely 1,000+ orders/day on Supabase Small or Medium compute. Redis comes after the queue, not before.

The single most valuable pre-launch test. Run k6 or Artillery against checkout for *one* variant with 10, 25 and 50 concurrent buyers. This is the one number in this document that cannot be estimated, and it decides whether Vokr oversells on its first busy day.

6. Production-Readiness Checklist

Derived from tracing the current codebase. For context on why each item exists, the relevant finding is noted. Items are grouped in section 7 (blocking), section 8 (recommended) and section 9 (deferred).

6.1 Current state of the codebase — what is real

Flow	Status	Finding
Every form on the site	Dead	All 62 forms across 27 pages are <code>onsubmit="return false;"</code> . No <code>fetch()</code> , no XHR, no API base URL anywhere in the codebase.
Cart	Mocked	In-memory JS array inside an IIFE; resets on navigation. Keyed by <i>display name string</i> — the PDP builds the variant key by concatenation, e.g. <code>"Model x (IN 9)"</code> . No SKU, no variant ID, no product ID anywhere.
Pricing	Client-controlled	Price read from a <code>data-price</code> DOM attribute. Live defect: the gift-card page offers ₹1000/2000/5000/10000 but the button is hardcoded <code>data-price="2000"</code> — selecting ₹10,000 adds ₹2,000.
Checkout	Absent	<code>alert('This is a demo checkout...')</code> . No checkout, cart, account or order page exists.
Auth	Cosmetic	No password field on the site. Any string in the email box flips the panel to "signed in". The Google button is <code>signIn('you@vokr.shop')</code> — a hardcoded literal.
Inventory	Nonexistent	Zero stock concept anywhere. No sold-out state, no availability, no disabled sizes. Sizes 4-11 render unconditionally.
Order tracking	Static copy	Inert form plus four descriptive <code><div></code> s. Not a tracker, not a state machine, no order lookup.
Search	Hardcoded, drifted	13-entry array duplicated across 27 files. Four pages carry 15 entries including <i>Model 251 Low</i> , which has no page.
Reviews	Fabricated	10 hardcoded objects with <code>verified: true</code> , a "4.7" average and a meta description claiming 4,059 reviews.
Images	Third-party	84 references to <code>cdn.shopify.com</code> . A variant homepage inlines 10 base64 images (five PNGs of 1.7-2.5 MB), producing a 19.9 MB page .
Tooling	None	No git, no <code>package.json</code> , no build step, no CI config, no <code>robots.txt</code> , no sitemap, no analytics.

6.2 Commitments already made in the legal copy

These are contractual, written into `terms.html` and the support pages, and the backend must honour or the copy must change.

Commitment	Implication
"order confirmation email/SMS with your order ID"	Must be amended to email — SMS is deferred (item R14).
COD on eligible orders and PIN codes, handling fee at checkout	Requires a courier. Razorpay does not do COD.
"courier serviceability at your PIN code — check at checkout"	Requires a serviceability data source.
COD refunds via bank transfer or Vokr store credit	Requires a credit ledger.
Returns: "no login required" — lookup by order number + email	Unauthenticated, rate-limited, enumeration-resistant endpoint.
Prices "inclusive of GST"	GST must be a computed per-variant field, never a constant — ₹295 laces and ₹9,995 shoes sit in different slabs.
Refunds in 5-7 business days; 6-month defect warranty; RTO re-ship charges; COD blacklist	Supporting entities and operational process.
Five published <code>@vokr.shop</code> addresses including statutory <code>grievance@</code>	Inbound mail required — Brevo sends only.

7. Required Before Launch

Nothing ships until every item below is true. Ordered by build sequence, not priority — all are blocking.

Foundation

#	Item	Detail
R1	Git repository with CI/CD	There is currently no version control. A payments system deployed by hand is not operable. GitHub Actions → Artifact Registry → Cloud Run.
R2	Artifact Registry cleanup policy	Keep 2 tags, delete untagged. 0.5 GB is the tightest quota in the stack. Set this in the first commit.
R3	Secrets in Secret Manager, loaded at boot	Exactly 6 versions: DB URL, Razorpay key ID, Razorpay key secret, Razorpay webhook secret, Brevo API key, Supabase service role key. Never fetch per request — 10,000 ops/month is ~13/hour.
R4	<code>max-instances=3</code> + billing budget alerts	Alerts at \$1 / \$5 / \$20. Free Tier is a discount, not a ceiling. This is the guardrail that prevents a four-figure surprise.
R5	Environment separation	Dev / staging / prod. Staging uses Razorpay sandbox credentials.

The money path

#	Item	Detail
R6	Server-side price resolution	The client never sends a price. Prices resolve from a variant ID, server-side. The gift-card defect is proof this failure mode is already live.
R7	Razorpay webhook signature verification	Order state is driven by the webhook, never by the browser callback.
R8	Idempotency keys	On order-creation and all payment endpoints. Prevents double-charging on retry or network blip.
R9	Inventory reservation inside a DB transaction	<code>SELECT ... FOR UPDATE</code> with a <code>CHECK (quantity_available >= 0)</code> constraint as the backstop. The Razorpay call must be outside the lock.
R10	Load test concurrent checkout on one variant	10 / 25 / 50 concurrent buyers of a single variant. LOAD TEST The number that decides whether Vokr oversells.
R11	GST computed per variant + compliant invoice	Sequential invoice numbering, HSN codes, state-wise CGST/SGST vs IGST. Confirm current footwear slabs with your CA.

Communications and fulfilment

#	Item	Detail
R12	Brevo configured as Supabase Auth custom SMTP	The most dangerous item on this list. The built-in mailer sends 2/hour and only to your own team — real customers get nothing, with no client-side error. Configure before any real signup exists.
R13	Raise the Supabase Auth email rate limit; forward client IP	Default is 30/hour after custom SMTP. Server-side Auth calls make every customer share Cloud Run's IP and one 30-per-5-minute bucket.
R14	Amend <code>terms.html</code> : "email/SMS" → "email"	Free, takes ten minutes, and forgetting it is a contractual mismatch on day one.
R15	Courier operational	Shiprocket account live with COD enabled and PIN serviceability available. Manual panel operation is acceptable at launch. Confirm plan tiers and whether API access is gated UNVERIFIED
R16	Razorpay live-mode KYC complete	Registered entity, PAN, GST, bank account. Test mode works immediately; live mode does not. Start early.

Data, security and content

#	Item	Detail
R17	Backups with a tested restore	6-hourly <code>pg_dump</code> to R2. Supabase Free has none. An untested backup is not a backup — perform a real restore before launch.
R18	Catalog cached in Cloud Run memory	~20 lines of code; quadruples Supabase egress headroom. Short TTL.
R19	HTTPS + HSTS; rate limiting; WAF enabled	Rate limits on auth, checkout and order-lookup endpoints. Cloudflare WAF and bot protection switched on.
R20	Remove the fabricated reviews	Delete the 10 hardcoded reviews, the "4.7" average and the "4,059 customer reviews" meta description. Publishing invented verified-purchase reviews is exposure under the Consumer Protection Act before it is an engineering problem.
R21	Homepage under 500 KB	The 19.9 MB base64 variant must not exist in production. Migrate all 84 images off <code>cdn.shopify.com</code> to R2, re-encoded as WebP/AVIF with responsive sizes.
R22	DPDP consent capture + log retention	Explicit consent records, data export and deletion endpoints, weekly Cloud Logging export to R2 to meet the one-year minimum retention.

8. Recommended but Not Blocking

Item	Rationale
Internal admin panel	Order list, status transitions, manual refund, inventory adjustment. Does not need to be attractive. Strongly recommended at MVP rather than a later phase — you cannot operate a store you cannot see into. Manual database access is a workable stopgap for the first days only.
Store credit / wallet ledger	Append-only, double-entry, never a mutable balance column. Needed by COD refunds and referrals; both are deferred, so this can follow them.
Order status email on every transition	Confirmed, shipped, delivered, return approved. Confirmation alone is the true blocker; the rest can land in the first week.
Audit log on orders and payments	Who changed what, when. Needed for dispute resolution and the Grievance Officer process referenced in the legal pages.
Structured request logging + uptime checks	Health endpoint and Razorpay webhook delivery monitoring, beyond Sentry error capture.
robots.txt, sitemap.xml, favicon, analytics	All absent from the current codebase.
Abandoned-cart and rate-limit row pruning	Protects the cumulative 500 MB Supabase storage limit. Uses the third Cloud Scheduler slot alongside reservation expiry.

9. Features Intentionally Deferred

Each of these was considered and consciously postponed. None is an oversight.

Feature	Deferred until	Reason
SMS / phone OTP	Order volume makes ₹5,900 trivial	DLT Principal Entity registration is ~₹5,900 one-time plus 3–7 days approval before a single message delivers. Requires the R14 copy change.
Redis	Postgres is measurably the bottleneck	Postgres covers rate limiting, cart and inventory locking. Adding Redis before measurement adds a vendor without evidence.
Background queue / worker	~300–500 orders/day	The first genuine architectural change. Triggered when inline email and courier calls start hurting checkout latency.
Search engine / Postgres FTS	~50+ SKUs	13 searchable entries do not justify either.
Real reviews	After orders exist	Gated on <code>order_item_id</code> so verified-purchase is enforced by a foreign key. Requires deleting the fabricated ones first (R20).
Wishlist, saved addresses, order history	Post-launch	Currently <code>href="#"</code> placeholders. No revenue dependency.
Referral program	After the credit ledger	The site promises ₹500 in Vokr credit; needs the ledger first.
Discount code engine	Post-launch	Implied by the subscription and discount-program pages.
Subscriptions	Last	<code>subscription.html</code> promises a fresh pair every 3/6/12 months at 15% off. Recurring debits in India require Razorpay Subscriptions with e-Mandate / UPI AutoPay, separate onboarding, RBI additional-factor authentication and mandatory 24-hour pre-debit notification. This is its own project, not a cron job.
Kids Model 123	Parental consent built	DPDP requires verifiable parental consent for under-18 data. Selling as an adult-purchased gift with no child account is fine; anything else waits.
Review photo uploads	With real reviews	Needs moderation queue.

10. Upgrade Path

Every trigger below is a **billing change, not an engineering one** — until the last row.

Trigger	Action	Cost	Notes
~70 orders/day, or any single day above ~75 orders	Brevo Starter	\$9/mo ~₹792	Removes the daily cap and Brevo branding. The first trigger you will hit.
Supabase egress crosses 3 GB, or ~250,000 pageviews/month	Supabase Pro	\$25/mo ~₹2,200	Removes the 402 cliff, adds daily backups with 7-day retention, removes the inactivity pause, raises DB to 8 GB and egress to 250 GB.
A third person needs the error dashboard	Sentry Team	\$26/mo ~₹2,288	Triggered by the 1-user cap, not by error volume.
~600+ orders/month	Shiprocket API plan	~₹499/mo UNVERIFIED	Manual order pushing becomes untenable. Confirm the tier that unlocks API access.
SMS becomes worth it	DLT registration + MSG91	₹5,900 one-time + ~₹0.15-0.25/SMS	Header and template fees, and whether renewal is annual, vary by operator UNVERIFIED .
Cold starts measurably hurt conversion	<code>min-instances=1</code>	~\$4-10/mo	Only after measurement. The keep-warm cron should make this unnecessary for a long time.
More than 5 WAF custom rules, or bot traffic affects revenue	Cloudflare Pro	\$20-25/mo	Not expected early.
~300-500 orders/day	Background job runner	Engineering time	The first real architectural change. Move outbound email and courier calls off the request path. Redis follows this, not before.

Perspective: at 2,000 orders/month the combined upgrades run roughly ₹3,000/month — about 0.15% of GMV at Vokr's price points. Infrastructure is not the cost problem at any point on this path. RTO on COD is.

11. Critical Architecture & Security Rules

Non-negotiable. Violating any of these reintroduces a defect the review already found, or breaks a limit the stack depends on.

Money and inventory

- **The client never sends a price.** Prices resolve server-side from a variant ID. Always.
- **Order state is driven by the Razorpay webhook**, never by the browser callback. Verify the signature on every webhook.
- **Idempotency keys on every payment and order-creation endpoint.**
- **Reserve inventory inside a database transaction** with `SELECT ... FOR UPDATE`, and add a `CHECK (quantity_available >= 0)` constraint as a backstop. Application logic alone is not sufficient.
- **Keep the Razorpay network call outside the row lock.** This single rule is worth more than any infrastructure choice for checkout throughput.
- **GST is a computed per-variant field**, never a constant.

Data and quota discipline

- **Never `SELECT *`**. Every column returned is Supabase egress against a 5 GB cliff that ends in HTTP 402.
- **No images and no logs in Supabase.** Images go to R2; logs go to Cloud Logging and then R2.
- **Cache the 6-SKU catalog in process memory** with a short TTL.
- **Prune abandoned carts and rate-limit rows on a schedule.** The 500 MB storage limit is cumulative and never resets.
- **Load secrets once at boot**, never per request.
- **Never exceed 6 active Secret Manager versions** — disable the old version when rotating rather than adding a seventh.

Security and privacy

- **Guest order lookup must be rate-limited and enumeration-resistant.** The returns page promises "no login required", which makes this endpoint unauthenticated by design.
- **Rate-limit auth, checkout and order-lookup** at both the application and Cloudflare layers.
- **Scrub PII before it reaches Sentry.** Addresses, phone numbers and emails must not land in error payloads.
- **DPDP:** explicit consent records, data export and deletion endpoints, and a one-year minimum retention for personal data, traffic data and processing logs. Substantive obligations become enforceable **13 May 2027**; build the consent architecture now and finish the rest before then.
- **Never publish fabricated reviews or ratings.**
- **Encryption at rest for addresses and phone numbers.**

Operational

- `max-instances` **is always set**. An unbounded instance count under a crawl loop is how free-tier accounts generate four-figure bills.
- **Billing budget alerts are always live.**
- **The keep-warm health check must touch Postgres**, not only Cloud Run — otherwise a quiet week could trigger Supabase's 7-day inactivity pause.
- **Restore drills are scheduled, not aspirational.**

12. Final Launch Decision

READY TO PROCEED

The stack is verified, internally consistent, and operable by a small team at approximately ₹250-400/month in fixed infrastructure. Nothing found in the review, the free-tier verification, the capacity analysis or the final sanity check justifies swapping a component. Every remaining issue is configuration or implementation discipline, not architecture.

Why READY

- Every component's free tier explicitly permits commercial production use and is available in the region Vokr needs.
- No component requires a paid plan for the functionality in scope at launch.
- Every failure mode is now known, and each has a mitigation already inside the stack.
- Removing Redis, queues, SMS and a search engine is safe and verified at this scale.
- The architecture supports 300-600 orders/day before any structural change is needed — far beyond launch requirements.

What READY does not mean

Ready to *build*, not ready to *launch*. The 22 blocking items in section 7 must all be true first. Two of them carry disproportionate risk:

- **R12 — Supabase Auth custom SMTP.** The only defect in this stack that fails *silently*, at zero traffic, on the very first real customer.
- **R10 — the concurrent-checkout load test.** The only number in this document that cannot be estimated, and the one that decides whether Vokr oversells on its first busy day.

Recommended build order

1. **Foundations** — git, Next.js scaffold, migrate the 27 pages into shared layouts and components, environments, Secret Manager, CI/CD, Sentry. This alone removes ~2 MB of duplication and the drifted search index.
2. **Catalog + auth** — schema and migrations, 6 SKUs seeded with real variant IDs and SKUs, Supabase Auth with Brevo SMTP, guest sessions. Delete the fabricated reviews here.
3. **Cart** — server-side, keyed by session or user, merged on login, prices resolved server-side. The gift-card defect dies here.
4. **Checkout and payments** — the highest-risk phase. Address capture, PIN validation, GST per line, Razorpay orders and webhooks, inventory reservation in a transaction. **Load-test before proceeding.**
5. **Orders and fulfilment** — guest lookup, Shiprocket, status emails, admin panel.
6. **Launch hardening** — WAF, rate limits, restore drill, DPDP consent, legal review of terms against what was actually built.

Standing decision to make now, not later. Pre-authorise the Supabase Pro upgrade (~₹2,200/month) so that whoever is on call can act the moment egress alerts fire, without needing a meeting. A 402 during the first festival sale would cost more than three years of Supabase Pro.

Document scope and limitations. Provider quotas and prices were verified on 7 September 2026 against official documentation and are subject to change with notice — re-verify before any major commitment. Items marked **UNVERIFIED** (Shiprocket plan tiers and API gating, `.shop` domain price, DLT header and renewal fees, Brevo free-plan branding on transactional email) must be confirmed by the team before they are relied upon. Items marked **LOAD TEST** cannot be resolved by research and require measurement of the real application. This document is an engineering reference; it is not legal, tax or financial advice. GST slabs, DPDP obligations and the terms and privacy copy should be reviewed by a qualified professional before launch.